

ARPIT TAGADE

Software & AI Engineering student

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PROFESSIONAL SUMMARY

Results-driven Software & AI Engineering student bridging the gap between intelligent algorithms and scalable system architecture. Specialized in full-stack development, ranging from low-latency WebSocket communication backends to highly interactive, AI-driven web applications. Proven ability to architect and deploy production-grade solutions utilizing Java Spring Boot, React, Next.js, and Google's Generative AI.

EDUCATION

Bachelor of Technology: Artificial Intelligence and Data Science,

Wainganga College of Engineering and Management

2029

Nagpur, Maharashtra

- Current First-Year Student
- Focus Areas: Algorithm Design, Data Structures, Programming Logic, AI Research, Full-Stack Architecture.

TECHNICAL ARSENAL

Languages

Java, C, C++, JavaScript/TypeScript, SQL

Backend & Systems

Spring Boot, Spring Security, Node.js, Express, WebSockets, REST API Development, JDBC, Oracle SQL

Core Competencies

Object-Oriented Programming (OOP), Multithreading, Real-Time Bidirectional Communication, State Management, UI/UX Engineering

Frontend Architecture

React.js, Next.js (App Router), Tailwind CSS, Framer Motion, HTML5/CSS3

Hardware & IoT

ESP32, Arduino Uno, Sensor Integration, Servo Control, OLED Interfaces, BIOS Optimization

ENGINEERING PROJECTS & ARCHITECTURE

Neosis: Real-Time Communication Platform, *Java Spring Boot, WebSockets, React, Tailwind CSS*

- Architected and deployed a full-stack, real-time chat application utilizing Spring Boot and WebSockets to ensure low-latency, bidirectional message passing.
- Engineered secure user authentication and session management pipelines utilizing Spring Security.
- Designed and built a highly responsive, modern frontend interface using React and Tailwind CSS, ensuring a seamless user experience across devices.

Intelligent Developer Portfolio & AI Assistant, *Next.js, TypeScript, Tailwind CSS, Gemini API*

- Architected a high-performance personal portfolio utilizing the Next.js App Router and Framer Motion for advanced, hardware-accelerated UI animations.
- Engineered a custom, real-time AI chat assistant integrated directly into the application state, powered by the ultra-fast gemini-2.5-flash model via Google's Generative AI SDK.
- Streamlined application delivery and continuous integration pipelines through automated Vercel deployments.

Monika-AI: Intelligent Web Interface, *Node.js, Vanilla JS, API Integration*

- Developed and deployed an interactive, server-backed web application providing users with an intelligent, responsive digital interface.
- Engineered the backend infrastructure using Node.js, ensuring rapid request handling and seamless API interactions.

Project CandyRobot: IoT & AI Integration, *ESP32, C++, Hardware Engineering, Cloud AI*

- Developed a dynamic, physical robotic interface combining ESP32 microcontrollers with servo motors and 128x64 OLED displays.
- Programmed the low-level hardware in C++ to interact seamlessly with Google's Gemini API over a network, creating an intelligent hardware system capable of real-time physical and visual feedback.

TECHNICAL INITIATIVES & LEADERSHIP

Founder, "Coding Challenge"

- Established and currently manage an educational platform delivering daily programming quizzes, logic challenges, and algorithmic problems to a growing community of software enthusiasts.

Hardware & Systems Optimization

- Demonstrated deep understanding of PC hardware architecture by diagnosing and manually optimizing BIOS-level RAM allocation and disabling resource-heavy background services to maximize compute and rendering performance.